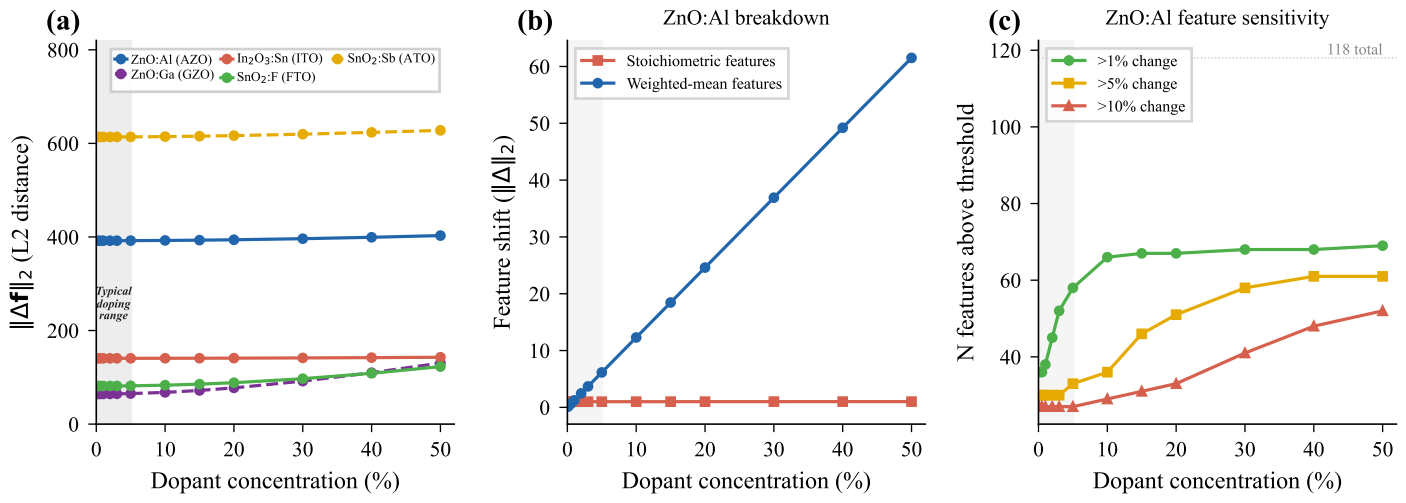
(a)
ZnO:Al (AZO)
In2O3:Sn (ITO)
SnO2:Sb (ATO)
ZnO:Ga (GZO)
SnO2:F (FTO)
Typical doping range
‖Δf‖2 (L2 distance)
Dopant concentration (%)
(b)
ZnO:Al breakdown
Stoichiometric features
Weighted-mean features
Feature shift (‖Δ‖2)
Dopant concentration (%)
(c)
ZnO:Al feature sensitivity
118 total
>1% change
>5% change
>10% change
N features above threshold
Dopant concentration (%)